Guess the Birthdate Trick

Ticking clocks and turning calendars

Grade 4

Introduction

With this birthdate-guessing activity, the teacher can correctly guess a student's birthdate. As students investigate how the trick works, they discover that any number can be written as a combination of powers of two. This is the same principle that computers use to represent numbers in binary.

Activity	Time	Description
Launch	5 min.	Ask students if they have ever seen a number-guessing magic trick. Invite a student. The teacher will demonstrate this activity by guessing the student's birth date. Show them the cards one by one and ask whether their birth date appears on each card. After they answer, reveal their date of birth. Repeat the trick with another student to build curiosity. Ask the class: How do you think this trick works? Template: https://tinyurl.com/birthdatetemplate
Try out the trick and explore	20 min.	After the demonstration, students can attempt the worksheet questions on the patterns they can find in the numbers on the cards. To do the trick, each pair of students can either cut out the magic cards from the template or make their own cards by writing the numbers on paper. They can find out each other's birth date with this trick. Student's Worksheet: https://tinyurl.com/birthdateworksheet
Discussion & Conclusion	10 min.	Discuss what patterns students noticed on the cards. Guide them to observe that the first number on each card is 1, 2, 4, 8, 16. Explain that these numbers double each time and are called powers of two.

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CT Components

Pattern Recognition
Observe and identify a common property of numbers and how they are grouped according to a shared structural rule.

Abstraction
Students focus on the important mathematical structure (powers of two) and ignore unnecessary details.

Algorithmic Thinking
We used a step-by-step approach of adding numbers on the cards in the top-left corner to identify the birth date.

Generalisation
Recognising a doubling pattern of 1, 2, 4, and so on, and adding different powers of 2 to make any number, we conclude that any number can be represented in terms of powers of 2.

Logic
Every card can correspond to a number

Activity Demonstration: Guess the Birth Date Trick
Ask students if they have ever seen a birthdate-guessing magic trick. The teacher will invite a student and will demonstrate this activity by guessing the student's birth date.

The teacher will show them the cards with dates written on them, one by one, and ask whether their birth date appears on each card.

Card A

1	3	5	7
9	11	13	15
17	19	21	23
25	27	29	31

Card B

2	3	6	7
10	11	14	15
18	19	22	23
26	27	30	31

Card C

4	5	6	7
12	13	14	15
20	21	22	23
28	29	30	31

Card D

8	9	10	11
12	13	14	15
24	25	26	27
28	29	30	31

Card E

16	17	18	19
20	21	22	23
24	25	26	27
28	29	30	31

When a student selects the card with their birth date, let's say, the selected cards are shown in the red dotted line.

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Card A

1	3	5	7
9	11	13	15
17	19	21	23
25	27	29	31

Card B

2	3	6	7
10	11	14	15
18	19	22	23
26	27	30	31

Card C

4	5	6	7
12	13	14	15
20	21	22	23
28	29	30	31

Card D

8	9	10	11
12	13	14	15
24	25	26	27
28	29	30	31

Card E

16	17	18	19
20	21	22	23
24	25	26	27
28	29	30	31

The trick is to add the values at the top left of the chosen cards: 1 + 2 + 16 = 19. The sum of these values will be the student's birth date. In this case, 19.

Another example: if the students choose cards B, C, and D, these 3 cards have a “2”, “4”, and “8” as their top-left numbers. If we add these numbers together, we get 14. 14 will be the student's birth date.

Card A

1	3	5	7
9	11	13	15
17	19	21	23
25	27	29	31

Card B

2	3	6	7
10	11	14	15
18	19	22	23
26	27	30	31

Card C

4	5	6	7
12	13	14	15
20	21	22	23
28	29	30	31

Card D

8	9	10	11
12	13	14	15
24	25	26	27
28	29	30	31

Card E

16	17	18	19
20	21	22	23
24	25	26	27
28	29	30	31

After the teacher has done a few demonstrations with other students, Students may think the trick works by memorizing numbers. The teacher needs to explain that the trick works because numbers follow a systematic pattern, which they will explore with this activity. Then, the teacher could ask students to look for patterns in the numbers on the cards.

Q. What patterns could you see on Card A?

a) All numbers are even
b) All numbers are odd
c) All numbers are multiples of 3
d) Numbers are not related to each other

Explanation

b) All numbers are odd

Numbers on Card A have 1,3,5,7….and so on. Therefore, all these numbers are odd. Also, if you notice numbers on card A have 1 in the 1’s place in the binary table.

CT Competency: Pattern Recognition

Q. What patterns could you see on Card E?

a) All numbers are even
b) All numbers are odd
c) All numbers are multiples of 3

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d) Numbers are 16 or greater

Explanation

d) Numbers are 16 or greater

Card E represents the 16's place in the binary table. Numbers appear here only if they include 16 in their representation. Therefore, all these Numbers are 16 or greater.

CT Competency: Pattern Recognition

Q. What patterns could you see on the Top-Left Numbers on all the cards?

a) All numbers are Powers of 2
b) All numbers are less than 10
c) All numbers are Powers of 3
d) Numbers are not related to each other

Explanation

b) All numbers are Powers of 2

These numbers double each time. They are called powers of two.

CT Competency: Pattern Recognition, Abstraction

Students may not understand why numbers appear on multiple cards. The teacher needs to explain that numbers are built from combinations of powers of two.

This trick may look like magic, but it is actually based on mathematics and logical rules. To understand it, let us try the trick out with your friend first.

Activity Making & Trail:

1. We will cut out the 5 cards given on the template. Or we can make cards ourselves by writing the numbers on paper as given in the template.

Card A			
1	3	5	7
9	11	13	15
17	19	21	23
25	27	29	31

Card B			
2	3	6	7
10	11	14	15
18	19	22	23
26	27	30	31

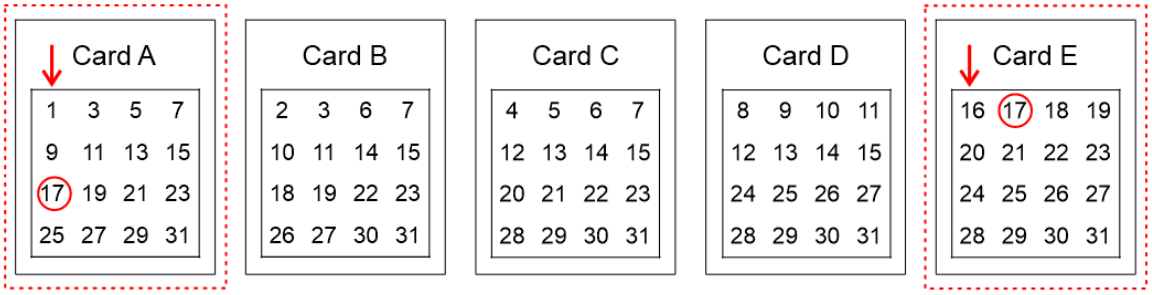
Card C			
4	5	6	7
12	13	14	15
20	21	22	23
28	29	30	31

Card D			
8	9	10	11
12	13	14	15
24	25	26	27
28	29	30	31

Card E			
16	17	18	19
20	21	22	23
24	25	26	27
28	29	30	31

2. We are now ready to do the trick. We will show our friend the 5 cards and ask them to select the cards with their birth date.
3. Now, we can figure out their birth date by simply adding up all the numbers in the top-left corner of the cards they selected.
4. For example, if our friend selected Card A and Card E, then their birth date would be 17, because the first numbers on these cards are 16 and 1, which add up to 17.

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Students may try to guess the number instead of using the rule. The teacher needs to emphasise that the trick follows a clear algorithm: add the top-left numbers of selected cards.

Q. What is the top-left number on Card D?

a) 2
b) 4
c) 8
d) 16

Explanation

c) 8
Card D has 8 in the top-left corner.
CT Competency: Pattern Recognition

Q. If the friend says their birth date appears on card B, card D, and card E. What is their birth date?

a) 24
b) 25
c) 26
d) 27

Explanation

c) 26
Add the top-left values of the cards: 16 + 8 + 2 = 26
CT Competency: Algorithmic Thinking, Abstraction

This activity shows that what first appears to be a magic trick is actually based on mathematics and logical rules, and a simple algorithm can be used to determine the birth date.

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